

SHAYAN BHOWMIK

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Introduction

Design, build, and ship production-grade AI systems and full-stack applications, spanning system architecture, RAG pipelines, and scalable backends. CSE undergraduate with strong fundamentals in applied ML and open-source engineering, building real-world systems through research and hackathons.

Education

B.Tech Computer Science and Engineering

Symbiosis Institute of Technology, Pune

2024 - 2028

GPA: 8.17 / 10

Experience

Kytes June 2026 – July 2026

Software Engineer Intern

Internship

- Architected an autonomous risk-response agent using LangGraph and OpenRouter, applying PMI/PMBOK strategies (Avoid, Transfer, Mitigate, Accept) to automate project risk decision-making.
- Engineered prompt frameworks and LLM orchestration logic for context-aware risk analysis, integrating a SQLite-backed pipeline for lightweight, low-latency data persistence.
- Collaborated with the engineering team to deploy the agent end-to-end, from backend architecture to production deployment on Streamlit Cloud.

Google

May 2026 – August 2026

Student Ambassador

Internship

- Represented Google Gemini on campus, driving developer-tool adoption among students through workshops, events, and technical content.

Projects

RAG Pipeline with Hybrid Search | FastAPI, Next.js, Supabase 2026 – Present

- Built a document ingestion pipeline for a hybrid RAG system, implementing semantic chunking, sentence-transformer embeddings, and BM25 sparse indexing over a document corpus.
- Designed the Supabase/pgvector schema and a unified ingestion endpoint to populate dense and sparse indexes end-to-end from raw documents in a single call.
- Currently extending the system with Reciprocal Rank Fusion (RRF), cross-encoder reranking, and Claude API-based grounded generation.

Deep Learning Model for Brain Tumor Detection | Python, PyTorch [\[Live Demo\]](#) Jan 2026 – Apr 2026

- Developed a custom CNN for binary brain tumor classification using a 2.5D representation, stacking consecutive axial slices from 786 patients across BRATS 2020 and OASIS to preserve inter-slice spatial context.
- Designed a patient-level evaluation pipeline with leakage-free splits, class-weighted loss, and probability aggregation with calibrated threshold selection, achieving >99% slice-level accuracy and 100% patient-level accuracy on the held-out test set (ROC-AUC 0.999).
- Applied Grad-CAM with brain-region masking for model interpretability and conducted ablation studies quantifying the contributions of 2.5D context, normalization, and class-weighted loss.

Skills

Languages: Python, C++, TypeScript

Frontend: React, Vite, Tailwind CSS, HTML, CSS

Backend: Node.js, Express.js, FastAPI, Flask

Databases: PostgreSQL, MySQL, Supabase

Tools & DevOps: Git, GitHub, Vercel, PyTorch, Postman, Linux, Docker

Soft Skills: Leadership, Problem-Solving, Ownership, Communication

Achievements

- Selected as **Google Student Ambassador** (2026), representing Google's developer programs on campus.
- Serving as **Research & Tech Head**, Symbiosis Quantum Club, leading quantum computing research initiatives.
- Reached the **Semi-Final round** of Smart India Hackathon (SIH) 2025, a national-level innovation competition.
- Participated in the **ISRO Bharatiya Antariksh Hackathon (BAH) 2026**, developing a space-tech solution.

Certifications

- **Claude Code 101** — Anthropic Jul 2026
- **Introduction to Generative AI** — Google Dec 2025
- **Introduction to Prompt Engineering** — Microsoft Dec 2025